

Material Safety Data Sheet

Product name

ZICOS PR-ET

1. Product and company identification

- a) Product Name ZICOS PR-ET
- b) Recommended use of the chemical and restrictions on use
- | | |
|---------------------------------|--|
| Recommended use of the chemical | Deodorant Agent; Skin-Conditioning Agent – Miscellaneous |
| restrictions on use | Not available |
- c) Manufacturer/Supplier/Distributor Information
- | | |
|------------------------|---|
| Name | ZIKO |
| Address | 688 Chokumro, Kumwangeup, Eumsunggun, Chungbuk, Korea |
| Emergency phone number | 82-43-877-3300 |

2. Hazards identification

- a) Hazard Classification
- Serious eye damage/eye irritation (Category 1)
- Acute aquatic hazard (Category 3)
- Long-term aquatic hazard (Category 3)
- b) Label elements including precautionary statements
- | | |
|-------------------------|--|
| Symbol |  |
| Single word | Danger |
| Hazard Statement | H318 Causes serious eye damage
H402 Harmful to aquatic life
H412 Harmful to aquatic life with long-lasting effects. |
| Precautionary Statement | |
| Precaution | P273 Avoid release to the environment.
P280 Wear protective gloves/eye protection/face protection.
P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310 Immediately call a POISON CENTER or doctor/physician. |
| Disposal | P501 Dispose of contents/container according to local, state and federal laws. |

3. Composition/Information on ingredients

- | | |
|------------------|-------------------------------------|
| Product name | ZICOS PR-ET |
| IUPAC name | 3-(2-Ethylhexyloxy)-1,2-propanediol |
| INCI name | Ethylhexyl Glycerin |
| CAS No. | 70445-33-9 |
| Concentration(%) | 100% |

4. First aid measures

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|-------------------|--|
| a) general advice | Consult a physician. Show this safety data sheet to the doctor in attendance. |
| b) Eye contact | Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. |
| c) Skin contact | Remove/Take off immediately all contaminated clothing. Gently wash with plenty of soap and water. |
| d) Inhalation | Remove victim to fresh air and keep at rest in a position comfortable for breathing. |
| e) Ingestion | Get medical advice/attention if you feel unwell. Rinse mouth. |

5. Fire-Fighting measures

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| a) Suitable (and unsuitable) extinguishing media | Use alcohol-resistant foam, dry chemical or carbon dioxide. |
| b) precautions for fire-fighters | Fire-extinguishing work is done from the windward and the suitable fire-extinguishing method according to the surrounding situation is used. Uninvolved persons should evacuate to a safe place. In case of fire in the surroundings: Remove movable containers if safe to do so. |
| c) Special protective equipment for firefighters | When extinguishing fire, be sure to wear personal protective equipment. |

6. Accidental release measures

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| a) Personal precautions, protective equipment and emergency procedures | Use personal protective equipment. Keep people away from and upwind of spill/leak.
Ensure adequate ventilation. Entry to non-involved personnel should be controlled around the leakage area by roping off, etc. |
| b) Environmental precautions and protective procedures | Prevent product from entering drains. |
| c) Methods and materials for containment and cleaning up | Absorb spilled material in a suitable absorbent. (e.g. rag, dry sand, earth, saw-dust)
In case of large amount of spillage, contain a spill by bunding. Adhered or collected material should be promptly disposed of, in accordance with appropriate laws and regulations. |

7. Handling and storage

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|---|--|
| a) Precautions for safe handling | Avoid contact with skin, eyes and clothing.
Handling is performed in a well ventilated place
Wear suitable protective equipment.
Prevent generation of vapour or mist. |
| b) Conditions for safe storage
(including any incompatibilities) | Wash hands and face thoroughly after handling.
Keep container tightly closed. Store in a cool and dark place.
Store away from incompatible materials such as oxidizing agents. |

8. Exposure controls & personal protection

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|--|---|
| a) Control parameters (e.g. occupational exposure limit values, biological limit values) | |
| Domestic provision | Not available |
| ACGIH provisions | Not available |
| Biological Exposure standard | Not available |
| b) Appropriate engineering controls | Install a closed system or local exhaust as possible so that workers should not be exposed directly. Also install safety shower and eye bath. |
| c) Personal protective equipment | |
| Respiratory protection | Vapor respirator. Follow local and national regulations. |
| Eye protection | Safety glasses. A face-shield, if the situation requires. |
| Hands protection | Protective gloves. |
| Body protection | Protective clothing. Protective boots, if the situation requires. |

9. Physical and chemical properties

a) Appearance	
physical state	Liquid(20℃)
Color	colourless
b) Odor	slightly
c) Odor threshold	Not available
d) pH	Not available
e) Meting point/freezing point	Not available
f) Initial boiling point and boiling range	125 °C(5torr)
g) Flashing point	152 °C
h) Evaporation rate	Not available
i) Flammability (solid, gas)	Not available
j) Upper/lower flammability or explosive limits	Not available
k) Vapor pressure	Not available
l) Solubility	Not available
m) Vapor density	Not available
n) Relative density	0.96
o) Partition coefficient: n-octanol/water	Not available
p) Auto-ignition temperature	Not available
q) Decomposition temperature	Not available
r) Viscosity	90~140mPas
s) Formula mass	204.31

10. Stability and reactivity

a) Chemical stability and possibility of hazardous reactions	Stable under recommended storage conditions.
b) Conditions to avoid	source of ignition
c) Incompatible materials	Oxidizing agents
d) Hazardous decomposition products	Carbon monoxide, Carbon oxides

11. Toxicological information

a) Information on the likely routes of exposure	Irritating to skin Irritating to eyes
b) Health hazards information	
Acute toxic	
oral	Not available
dermal	Not available
inhalation	Not available
Skin corrosive/irritant	Not available
Serious eye damage/eye irritation	Not available
Respiratory sensitization	Not available
Skin sensitization	Not available
Carcinogenicity	
IARC	Not available
OSHA	Not available
ACGIH	Not available
NTP	Not available
EU CLP	Not available
Germ Cell Mutagenicity	Not available
Reproductive toxicity	Not available
Specific target organ toxicity (single exposure)	Not available
Specific target organ toxicity (repeated exposure)	Not available
Aspiration hazard	Not available

12. Ecological information

a) Aquatic and terrestrial ecotoxicity	
Fish	Not available
Crustacea	Not available
Algae	Not available
b) Persistence and degradability	
Remaining characteristic	Not available
Resolvability	Not available
c) Bioaccumulative potential	
Biodegradation	Not available
d) Mobility in soil	Not available
e) EU classification information (Confirmed classification results)	Xi: R41, R52 R53
f) EU classification information(Danger phrases)	R41, R52, R53
g) EU information classification(safety phrases)	S:(2)-26-39-61
h) Other adverse effects	Not available

13. Disposal considerations

a) Disposal method	
Product	Recycle to process, if possible. Consult your local regional authorities. You may be able to burn in a chemical incinerator equipped with an afterburner and scrubber system.
b) Disposal precaution	Dispose of contents / container in accordance with local regulation.

14. Transport information

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|--|---|
| a) UN number(UN No.) | ARD/RID : Not applicable
IMDG : Not applicable
IATA : Not applicable |
| b) UN proper shipping name | ARD/RID : Not dangerous goods
IMDG : Not dangerous goods
IATA : Not dangerous goods |
| c) Transport hazard class | ARD/RID : Not applicable
IMDG : Not applicable
IATA : Not applicable |
| d) Packing group (if applicable) | ARD/RID : Not applicable
IMDG : Not applicable
IATA : Not applicable |
| e) Environmental hazards | ARD/RID : Not applicable
IMDG Marine pollutant : Not applicable
IATA : Not applicable |
| f) Special precaution which a user to be aware of or needs to comply with in connection with transport or conveyance either within or outside their premises | |
| Type of fire emergency action | Not applicable |
| Type of exposure emergency actions | Not applicable |

15. Regulatory information

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|---|----------------|
| a) Industrial Safety and Health Act | Not applicable |
| b) Toxic Chemical Control Act | Not applicable |
| c) Dangerous Material Safety Control Act | Not applicable |
| d) Wastes Management Act | Not applicable |
| e) Other requirements in domestic and other countries | Not applicable |

16. Other information

- | | |
|---|------------|
| a) Information source and references | |
| TCI(f) Initial boiling point and boiling range) | |
| TCI(g) Flashing point) | |
| b) Issuing date | 2017-02-16 |
| c) Revision number and date | |
| Revision number | 0 |
| Revision date | 0 |
| d) others | |

- ☐ This SDS was prepared sincerely on the basis of the information we could obtained, however, any warranty shall not be given regarding the data contained and the assessment of hazards and toxicity. Prior to use, please investigate not only the hazards and toxicity information but also the laws and regulations of the organization, area and country where the products are to be used, which shall be given the first priority.

The products are supposed to be used promptly after purchase in consideration of safety. Some new information or amendments may be added afterwards. If the products are to be used far behind the expected time of use or you have any questions, please feel free to contact us. The stated cautions are for normal handling only. In case of special handling, sufficient care should be taken, in addition to the safety measures suitable for the situation. All chemical products should be treated with the recognition of "having unknown hazards and toxicity", which differ greatly depending on the conditions and handling when in use and/or the conditions and duration of storage. The products must be handled only by those who are familiar with specialized knowledge and have experience or under the guidance of those specialists throughout use from opening to storage and disposal.

Safe usage conditions shall be set up on each user's own responsibility.